

Kharka Farm OS — Mobile work summary

Project: Livestock monorepo · App: `apps/mobile` (Expo / React Native)
Date: Sunday, 20 September 2026 · Branch context: feature work on `app` / local mobile parity

Today's goal: Stabilize and harden the native mobile app — fix runtime API/task errors, improve scan & search UX, audit every screen for real bugs, and ship concrete fixes across shed rounds, inbox deep-links, animals, breeding, health, revenue, and related modules.

1. Database & API — task enum crash

DATABASE Nest API was spamming `GET /v1/tasks` failures: *Value 'ANESTRUS_VET' not found in enum 'TaskType'*.

- Inspected Postgres `TaskType` enum vs Prisma schema.
- Remapped legacy rows: `ANESTRUS_VET`, `ANESTRUS_MINERAL`, `ANESTRUS_DECISION`, and other orphan labels → current schema values (e.g. `SILENT_HEAT_CHECK`, `REPEAT_BREEDER`).
- Added migration `20260920143000_remap_legacy_task_types` and a one-off remap script.
- Verified `prisma.task.findMany()` and smoke-tested mobile-facing endpoints (all 200).

2. Scan & search UX

- FEATURE** **Scan hub** — live animal/batch suggestions while typing (280ms debounce), not only on “Find”.
- FIX** UUID / QR payloads resolve via API instead of inventing a fake short tag.
- FEATURE** **AnimalIdSearch** — same live typeahead + clear network vs empty-state messaging.
- FIX** Scan overlay no longer advances with a bogus tag when detail fetch fails.
- UI** QR **scan FAB icon** redesigned: viewfinder corners + QR modules + scan beam (was ID-card glyph).

3. Inbox deep-links & module routing

Task `actionPath` values like `/breeding?form=heat&animalId=...` were ignored on mobile (opened Breeding/Health with no tab or animal).

- Added `lib/action-path.ts` + unit tests.
- Wired Inbox open-task → parsed stack destinations.
- Breeding / Health / Feed / Revenue honor route params (`form`, `type`, `animalId`, `breedingId`).
- Breeding PD / calving / colostrum: selecting a record switches tab and shows selection; required animal checks on save.

4. Shed rounds & animal detail (critical bugs)

Issue	Fix
Health-check / marker rounds fell through to milk save	Mode-specific UI + <code>saveHealthLocal</code> ; unsupported modes blocked

Issue	Fix
Successful save always showed "offline"	Clear error on success
Undo visible in all modes but only undid milk	Undo only for milking
Resume round failed silently	Clear server id + user-facing error
Animal detail kept previous animal on id change	Reset cache/state and reload
<code>showBack</code> on AppShell ignored	Back control implemented
Empty milk/weight submitted as 0	Client-side numeric validation

5. Second-pass audit fixes (remaining screens)

From the remaining-screens audit follow-up:

- **CRASH** Dashboard `data?.alerts.healthOverdue` → safe `alerts?` chaining.
- **DATA** Expenses only enqueue offline on network/5xx — not on validation 4xx.
- **UX** Batch detail resets when route `id` changes.
- **UX** Revenue coop payment moved to FlatList footer; mark-paid requires confirm.
- **UX** Tank delivery: receipt upload failure reported separately after delivery save.
- Tasks module-cache `dueBefore` computed at prefetch time (not frozen at import).

6. Verification

- Mobile `pnpm typecheck` — clean.
- Mobile unit tests — 20/20 passed (including new action-path tests).
- API smoke script — login + dashboard, tasks, animals, search, batches, breeding, health, production, feed, inventory, revenue, expenses, rounds — all OK.

7. Key files touched (representative)

- `apps/mobile/src/components/ModuleIcon.tsx` — new QR scan icon
- `apps/mobile/src/screens/ScanHubScreen.tsx`, `AnimalIdSearch.tsx`, `ScanAnywhere.tsx`
- `apps/mobile/src/lib/action-path.ts`, `InboxScreen.tsx`, `BreedingScreen.tsx`, `HealthScreen.tsx`
- `apps/mobile/src/screens/RoundScreen.tsx`, `HomeScreen.tsx`, `AnimalDetailScreen.tsx`, `AppShell.tsx`
- `DashboardScreen.tsx`, `ExpensesScreen.tsx`, `BatchDetailScreen.tsx`, `RevenueScreen.tsx`, `TankDeliveryScreen.tsx`
- `apps/api/prisma/migrations/20260920143000_remap_legacy_task_types/`

8. Suggested next steps (not done today)

- Offline queue for animal *edit*, inbox snooze/dismiss.
- PnL / reports cache keys per period; numeric validation pass on Production/Feed/Inventory forms.
- Conflicts screen richer compare UI; print error handling on Daily sheet / Tags.

- Device QA pass on emulator + physical phone with `EXPO_PUBLIC_API_URL` .

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This document summarizes agent + developer work in the Cursor session on the mobile app hardening pass.